COURSE CODE: SECR3253

NETWORK PROGRAMMING

STUDENT NAMES & MATRIC:

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SECTION: 2

My Contribution to the Project

As a member of the Nahj Net Automation project, I was responsible for developing the Linux Activity component using Ansible automation. My main task was to create an Ansible playbook that automates the collection of Linux system activity and information from remote hosts. This contribution was important because it reduced the need for manual system monitoring and ensured that administrators could retrieve system information efficiently and consistently.

To complete this task, I first studied the overall project structure to understand how the repository was organized and how different playbooks interacted with each other. Since the project follows an Infrastructure as Code (IaC) approach, it was important that my work followed the same coding style and directory structure used by the rest of the team.

After understanding the project requirements, I developed the linux_activity.yml playbook using YAML syntax and Ansible modules. The playbook was designed to execute commands on remote Linux machines through SSH and collect relevant system information. Throughout development, I continuously tested the playbook to ensure that it executed correctly without errors and produced the expected results. Whenever problems occurred, I reviewed the error messages, corrected the syntax, and tested the playbook again until it worked successfully.

Besides writing the playbook, I also became involved in the team's version control workflow using Git and GitHub. I created my own feature branch, worked independently without affecting the main branch, and learned how to commit and push my changes. This workflow allowed every team member to work on separate tasks while keeping the project organized. Through this experience, I gained a better understanding of collaborative software development and how modern development teams manage code changes.

Overall, I successfully completed my assigned feature while ensuring that it could be integrated smoothly into the project repository. My contribution supported the project's objective of automating network and system administration tasks using Ansible.

Challenges Faced During the Project

This project presented several challenges that helped me improve my technical and problem-solving skills. One of the main challenges was learning Ansible because it was a new technology for me. Although I had some programming experience, automation using playbooks was different from traditional programming. I needed to understand concepts such as inventories, hosts, tasks, modules, variables, and YAML syntax before I could build a working playbook.

Another challenge was dealing with YAML formatting. Unlike many programming languages, YAML depends heavily on correct indentation. A single misplaced space or incorrect indentation level could cause the playbook to fail. At first, it was difficult to identify these formatting mistakes, but after carefully reading error messages and testing repeatedly, I became more familiar with the syntax and learned how to avoid these issues.

Testing the playbook also required patience. Sometimes commands did not execute as expected because of inventory configuration, SSH connectivity, or module usage. I spent time reviewing Ansible documentation and experimenting with different approaches until I found the correct solution. This trial-and-error process improved my troubleshooting skills and taught me the importance of systematic debugging.

Another challenge involved using Git and GitHub for collaborative development. Before this project, I had limited practical experience with Git workflows. I needed to learn how to clone repositories, create branches, switch between branches, stage files, write meaningful commit messages, push changes to GitHub, and understand pull requests. Initially, these commands seemed confusing, but with practice I became more confident using Git as part of the software development process.

Time management was another challenge. Since multiple project tasks had to be completed within a limited period, I needed to organize my work carefully. I divided the task into smaller steps, including understanding the requirements, implementing the playbook, testing, debugging, and documenting my work. This approach helped me complete my assigned feature more effectively.

Despite these challenges, overcoming each obstacle increased my confidence and improved my technical abilities. Every problem I solved became an opportunity to learn something new.

What I Learned from the Project

This project provided valuable practical experience beyond what I learned in lectures. One of the most important lessons was understanding how automation can simplify system administration. Instead of manually logging into multiple Linux servers and executing commands one by one, Ansible allows administrators to automate these repetitive tasks through playbooks. This demonstrated how automation improves efficiency, consistency, and reliability in real-world IT environments.